(P2) Starting from (4.91)

$$Z(N,T,P) = \int_0^\infty dV e^{-\beta PV} \int \frac{1}{N!} \prod_{i=1}^{N} \frac{d^3\vec{q}_i\, d^3\vec{p}_i}{h^3} \exp\left\{-\beta \sum_{i=1}^{N} \frac{p_i^2}{2m}\right\}$$

Integral over $\frac{d^3\vec{q}_i}{h^3} = \frac{V^N}{h^{3N}}$

The rest of the second integral is exactly the same, with N copies.

$$Z_1 = \int d^3\vec{p}_i\, e^{-\frac{\beta p_i^2}{2m}}$$

$$Z(N,T,P) = \int_0^\infty \frac{V^N}{N!\, h^{3N}} dV e^{-\beta PV} Z_1^N$$

$$Z_1 = \left[\int dp\, e^{-\frac{\beta p}{2m}}\right]^3$$ Power of 3 for 3 degrees of freedom in momentum

$$\int dp\, e^{-\frac{\beta p}{2m}} = \sqrt{\frac{2\pi m}{\beta}} = \sqrt{2\pi m k T}$$

So now $\left(\frac{\sqrt{2\pi m k T}}{h}\right)^{3N} = \left(\frac{1}{\lambda(T)}\right)^{3N}$ $\quad \lambda(T) = \frac{h}{\sqrt{2\pi m k T}}$

$$Z = \int_0^\infty dV V^N e^{-\beta PV} \frac{1}{N!\, \lambda(T)^{3N}}$$ let $u = \beta PV$ $\quad du = \beta P dV$, $V = \frac{u}{\beta P}$

$$\frac{1}{(\beta P)^{N+1}} \int e^{-u} u^N du = \frac{N!}{(\beta P)^{N+1}}$$ so $Z = \frac{1}{(\beta P)^{N+1} \lambda(T)^{3N}}$

$$G = -kT \ln Z = -kT \ln\left(\frac{1}{(\beta P)^{N+1}} \frac{1}{\lambda^3}\right)$$

$$= -kT\left((N+1)\left(\ln\frac{1}{\beta} + \ln\frac{1}{P}\right) - 3\ln\lambda\right)$$

$$= NkT\left[-\ln kT + \ln P + 3\ln\left(\frac{h}{\sqrt{2\pi m k T}}\right)\right]$$

$$= NkT\left[-\ln(kT) + \ln P + 3\left(-\tfrac{1}{2}\right)\ln kT + 3\ln\left(\frac{h}{\sqrt{2\pi m}}\right)\right]$$

$$= NkT\left[\ln P - \tfrac{5}{2}\ln kT + \tfrac{3}{2}\ln\left(\frac{h^2}{2\pi m}\right)\right. \qquad (4.92)$$

For 4.93

$$dG = -SdT + VdP + \mu dN$$

$$\left(\frac{dG}{dP}\right)_{T,N} = -S\frac{dT}{dP} + V + \mu\frac{dN}{dP} = V$$

Using (4.92) $\frac{dG}{dP} = \frac{NkT}{P} = V \Rightarrow PV = NkT$ (4.93)

(P3) Using (4.95) $Z(N,T,B) = [2\cosh(\beta\mu_0 B)]^N$ (Ising Model)

Then $G = -kT\ln Z = -NkT\ln[2\cosh(\beta\mu_0 B)]$ (4.96)

$$M = -\frac{\partial G}{\partial B} = \cancel{NkT}\,\mu_0\cancel{\beta}\frac{2\sinh(\beta\mu_0 B)}{2\cosh(\beta\mu_0 B)} = N\mu_0\tanh(\beta\mu_0 B)$$

Expand for small B. $\tanh(x) \sim x$ for small x

So $M \simeq N\mu_0\beta\mu_0 B = \frac{N\mu_0^2}{kT}B$

Then $\chi = \left.\frac{\partial M}{\partial B}\right|_{B=0} = \frac{N\mu_0^2}{kT}$ (4.98)

$\boxed{P4}$ a) $Z(T,B) = \sum_{\{s\}} e^{\beta B M_z} = \sum_{\{s\}} e^{\beta B \mu \sum m_i}$

$$Z = \sum_{\{s\}} e^{\beta B \mu m_1} e^{\beta B \mu m_2} \cdots e^{\beta B \mu m_N}$$

Look at one m_0 value. It can have values from $-s$ to s

$$\sum_{\{s\}} e^{\beta B \mu m} = e^{\beta B \mu (-s)} + e^{\beta B \mu (-s+1)} + \cdots e^{\beta B \mu (s-1)} + e^{\beta B \mu s}$$

Pairs of matching terms. Could be $s=0$ term with no pair, but thats ok as we'll see.

So $\sum_{\{s\}} e^{\beta B \mu m} = \left(2\cosh(\beta B \mu s) + 2\cosh(\beta B \mu (s-1)) + \cdots\right)$

$$= \left(2 \sum_{i=0}^{s} \cosh(\beta B \mu i)\right)$$

Total Z then: $Z = \left(2 \sum_{i=0}^{s} \cosh(\beta B \mu i)\right)^N$

b) $G = -kT \ln Z$ For small B $\cosh \beta B \mu i \simeq 1 + \frac{(\beta B \mu i)^2}{2}$

$$G = -NkT \ln\left[2 \sum_{i=0}^{s} \left(1 + \frac{(\beta B \mu i)^2}{2}\right)\right] = -NkT \ln\left[2(s+1) + \sum_{i=1}^{s} (\beta B \mu i)^2\right]$$

$$G = -NkT \ln\left[2s+1 + \beta^2 B^2 \mu^2 \frac{s(s+1)(2s+1)}{6}\right]$$

$$= -NkT \ln\left[(2s+1)\left(1 + \frac{B^2 \mu^2}{6k^2T^2} s(s+1)\right)\right]$$

$$= -NkT\left(\ln[2s+1] + \ln\left[1 + \frac{B^2\mu^2}{6k^2T^2} s(s+1)\right]\right)$$

let $-NkT \ln(2s+1) = G(0)$

Then $G = G(0) - NkT \ln\left(1 + \frac{B^2\mu^2}{6k^2T^2} s(s+1)\right)$

(P4) cont..

Now using $\ln(1+x) \simeq x$ when x is small

$$G = G(0) - \frac{N\mu^2 s(s+1) B^2}{6kT} + O(B^4)$$

next term in $\ln(1+x)$ expansion is $-\frac{x^2}{2} \sim B^4$

c) $\chi = \frac{\partial M}{\partial B}\Big|_{B=0}$ $\qquad M = -\frac{\partial G}{\partial B} = \frac{N\mu^2 s(s+1) B}{3kT}$

$$\frac{\partial M}{\partial B}\Big|_{B=0} = \frac{N\mu^2 s(s+1)}{3kT} \qquad \text{let } c = \frac{N\mu^2 s(s+1)}{3k}$$

Then $\chi = \frac{c}{T}$

d) $C_B = \left(\frac{\partial H}{\partial T}\right)_B$ $\qquad H = -\frac{\partial \ln Z}{\partial \beta}$ $\qquad \ln Z = -\frac{G}{kT} = -\beta G$

$$H = \frac{\partial \ln(\beta G)}{\partial \beta} = G + \beta \frac{\partial G}{\partial \beta} \qquad \frac{\partial G}{\partial \beta} = \frac{N B^2 \mu^2 s(s+1)}{6}$$

$$H = G(0) - \frac{N\mu^2 B^2 s(s+1)}{3}\beta = G(0) - \frac{N\mu^2 B^2 s(s+1)}{3kT}$$

$$C_B = \left(\frac{\partial H}{\partial T}\right)_B = \frac{N\mu^2 B^2 s(s+1)}{3kT^2}$$

For C_M, write G in terms of M: $\qquad B = \frac{3kTM}{N\mu^2 s(s+1)}$

$$G = -\frac{9}{6}\frac{kTM^2}{N\mu^2 s(s+1)} = \frac{-TM^2}{2c} \quad \text{(ignoring } G(0)\text{)}$$

$$H = G + \beta\frac{\partial G}{\partial \beta} = G + \beta\frac{\partial}{\partial \beta}\left(\frac{-M^2}{2ck\beta}\right) = G + \beta\left(\frac{M^2}{2ck\beta^2}\right)$$

$$H = G + \frac{M^2}{2c\beta k} = -\frac{T}{2c}M^2 + \frac{T}{2c}M^2 = 0$$

So $C_M = \left(\frac{\partial H}{\partial T}\right)_M = 0$ and $C_B - C_M = \frac{cB^2}{T^2}$